

Math 300

Quiz # 7

I will not cheat on this quiz. _____

1. (2 pts) Give examples of two **finite** sets with different cardinalities.

$$A = \{1, 2\}$$

$$B = \{1, 2, 3\}$$

2. (2 pts) Give examples of two **infinite** sets with different cardinalities.

$\mathbb{N}$ countably infinite

$\mathbb{R}$ uncountable

3. (4 pts) Consider the closed interval $[0, 5]$.

a) Give an example of a partition, P , of this interval into 5 subintervals if the width of your largest subinterval is 2.

$$P = \{0, 2, 3, 4, 4.5, 5\}$$

b) Give a refinement, Q , of your partition in part a.

$$Q = \{0, 1, 2, 3, 4, 4.5, 5\}$$

4. (2 pts) Let $A = \left\{1 + \frac{1}{n} \mid n \in \mathbb{N}\right\}$. What are the supremum and the infimum of A ?

$$\sup(A) = 2$$

$$\inf(A) = 1$$

OVER $\rightarrow$

5. (2 pts) Give an example of an uncountable set that is bounded above but does not contain its supremum.

Any interval of real numbers of the form $[a, b)$
or (a, b) , must be open on the upper side.

6. (2 pts) Identify the first few decimal digits of a real number b that could result from applying Cantor's second diagonalization method to the sequence of real numbers given below.

$$a_1 = 0.1010999\dots$$

$$a_2 = 0.0123456\dots$$

$$a_3 = 0.3333333\dots$$

$$a_4 = 0.5656565\dots$$

$$a_5 = 0.2500000\dots$$

$$a_6 = 0.1111111\dots$$

or

$$b = 0.000010$$

$$b = 0.001110$$

$$b_i = 0 \text{ if } a_{ii} \neq 0$$

$$b_i = 1 \text{ if } a_{ii} = 0$$

$$b_i = 0 \text{ if } a_{ii} = 1$$

$$b_i = 1 \text{ if } a_{ii} \neq 1$$

7 (2pts) The first few sets in a sequence defined by a one-to-one correspondence, g , mapping the natural numbers N to the power set $P(N)$ are shown in the list below. Identify the first few elements of the set D that Cantor would use to show that this function g cannot map onto all of $P(N)$.

$$g(1) = \{2\}$$

$$g(2) = \{1, 2, 4, 8, 16, 32, \dots\}$$

$$g(3) = \{3\}$$

$$g(4) = \{1, 2, 3\}$$

$$g(5) = \{1, 3, 5, 7\}$$

$$g(6) = \{\}$$

$$D = \{1, 4, 6, \dots\}$$

8. (4 pts) List four topics that you enjoyed learning about in this course.

I did not cheat on this quiz. _____